Sectoral Correlation Networks of the Philippine Stock Market: A Minimum Spanning Tree Approach

Jemica Albania Jerusalem Cruz Jaizel Dela Paz William James Landicho
Franxine Reyes Luis Paul Valencia Lorenz Gerald Yambao

BSM CS3A - Group 6

Abstract

This paper examines the network structure and stability of the Philippine Stock Exchange (PSE) during various economic periods namely from pre-pandemic to Covid-19 pandemic downturn. Based on Minimum Spanning Tree (MST) technique, the study constructs six key sectors including Financials, Industrial, Holding Firms, Property, Services, and Mining and Oil to analyse the proximate distance among stocks. Data records were classified as two sets: Batch-A (July 2019 to November 2019) for a stable pre-pandemic market, and Batch-B (December 2019 to May 2020) for a volatile market at the beginning of the pandemic. The results show a dramatic decrease of the market's MST in the pandemic, when volatility and correlations among stocks were escalating. The Mining and Oil and Financial sectors had the biggest drops, signaling the severe uncertainty and economic strain during the crisis. The results give an understanding for fragility of the Philippine financial market and systemic risks in the global health crisis.

INTRODUCTION

An Increase in Systemic Risk in the financial system threatens the financial stability. It is defined as the risk of a serious failure occurring in a part or whole of the financial system that is undesirable for the growth of economy or an increase in the vulnerability of the financial system (Hodula et al., 2022).

There are two dimensions in Systemic Risks in any financial system a cyclical dimension that concerns with the macro-financial imbalances over the financial cycle, while the structural dimension is concerned with the build up of risks caused by the changes in the financial system(Hodula et al., 2022).

The Covid-19 pandemic and the subsequent Enhanced Community Quarantine has caused grave concern and volatility within the financial markets the Philippine Stock Exchange Index (PSEi) declined to an eight-year low of $4,623.42$ points in 19 of March from $7,815.26$ of 31 December of 2019. A temporary increase in lending premiums was observed during that period as the Philippines 10-year bond yields had a slight increase of 5 percent and eased to 4.5percent but it remained elevated (Bangko Sentral ng Pilipinas, 2023).

The first Use of correlation-based networks using graph theory tools was in 1999 by Professor Rosario Mantegna. Where they introduced the usage of minimum spanning tress to model the relationship of United States stocks relative to their industry sectors and reflects the price of the stock beyond historical and current financial standing of the company but also information about its structure and topology (Danko & Šoltés, 2018). Minimum spanning tress will be the fundamental component of this exposition as it will be used to describe the structural relationships and potential drastic developments of the tree during the distressing financial market reactions to Covid-19 and ECQ.

OBJECTIVES

The main objective of the exposition paper is to implement a minimum spanning tree graph abstracted for the six macro-economic aggregate of the Philippine Stock Exchange or its sectors. This facilitates an exploration on the correlations the sectors had with each other before and during the Covid-19 pandemic that shows the structural risks attained by the drastic change in the financial system it produces. Our main objectives are as follows:

1. Select pair of sectors that their correlations are highly different from 2019 stable market and 2020 pandemic unstable market.
2. Identify the stable sector pair connections across market regimes that constitute the structural backbone of the sectoral network.
3. Evaluate how market regime shifts affect the distance structure of sector correlations using the Minimum Spanning Tree representation.

PRELIMINARY

Mantegna's study on Market Structures

The paper will utilize the 1999 paper of Mantegna in constructing the market structure with MST of the PSEI Composite with some modifications to scale down the technical load associated with reproducing the original paper such as:

Mantegna (1999). Specified that on their data sets they took large samples of the Standard and Poor's 500 index specifying that 443 stocks was picked as the nodes of the MST. This is a very large data sources for an expository paper and the stocks was abstracted towards several economic sectors that is presented by the Philippine Stock Exchange.

Philippine Stock Exchange Sectors

Philippine Stock Exchange (PSE) is subdivided to six sectors of stocks these are are: Financials, Industrial, Holding Firms, Property, Services, Mining and Oil. each sector is composed of the companies listed on PSE that is from that sector (The Philippine Stock Exchange, Inc., 2026). This six sectors will be the data sets to be modeled.

Time Frame of the Data

Prior to the Covid-19 pandemic the financial markets was hit during the market condition shift to the pandemic economic dowturn during the December of 2019 towards May of 2020 the six sectors of the PSE generally exhibited substantial downturns with Mining and Oil followed by financials showed the most aggregate decline during the time frame. this timeframe is also an extremely highly volatile point in the market with high uncertainty (Cacnio, 2020). This time frame December 2019 to May 2020 will be the pandemic market and the sable pre-pandemic market as July 2019 to November 2019 the details are in 1.

Table 1

Schedule Comparison for Weekly Intervals by Batch

Batch	**Start Date**	**End Date**
Batch A	7-1-2019	11-25-2019
Batch B	12-2-2019	5-25-2020

Baseline Graph

As the study aimed to measure the correlation of one sector towards the other a K6 complete weighted graph will be the starting graph to show the relationships each sectors had with the others as shown in 1 and formally defined such that: $K_6 = (V, E)$, where V are the vertices representing the six PSE sector while E are the edges representing the weight (quantified in correlation coefficient or distance dependent on the stated graph) between two vertices or sectors

Beyond the Baseline Graph

The original paper utilized the correlation coefficient as the fundamental unit determining similarity between the nodes the formula utilized logarithmic rate of return as its predictors. Then transferred to a series of correlation coefficients over series of time horizons to detect hierarchical structures between the nodes, then a euclidean metric was utilized to determine the relative distances of the nodes with one another in a distance matrix to create the MST (Mantegna, 1999). the paper is expected to exhibit these graphs:

1. K6 complete graph where edge weights is the batch a correlation coefficient

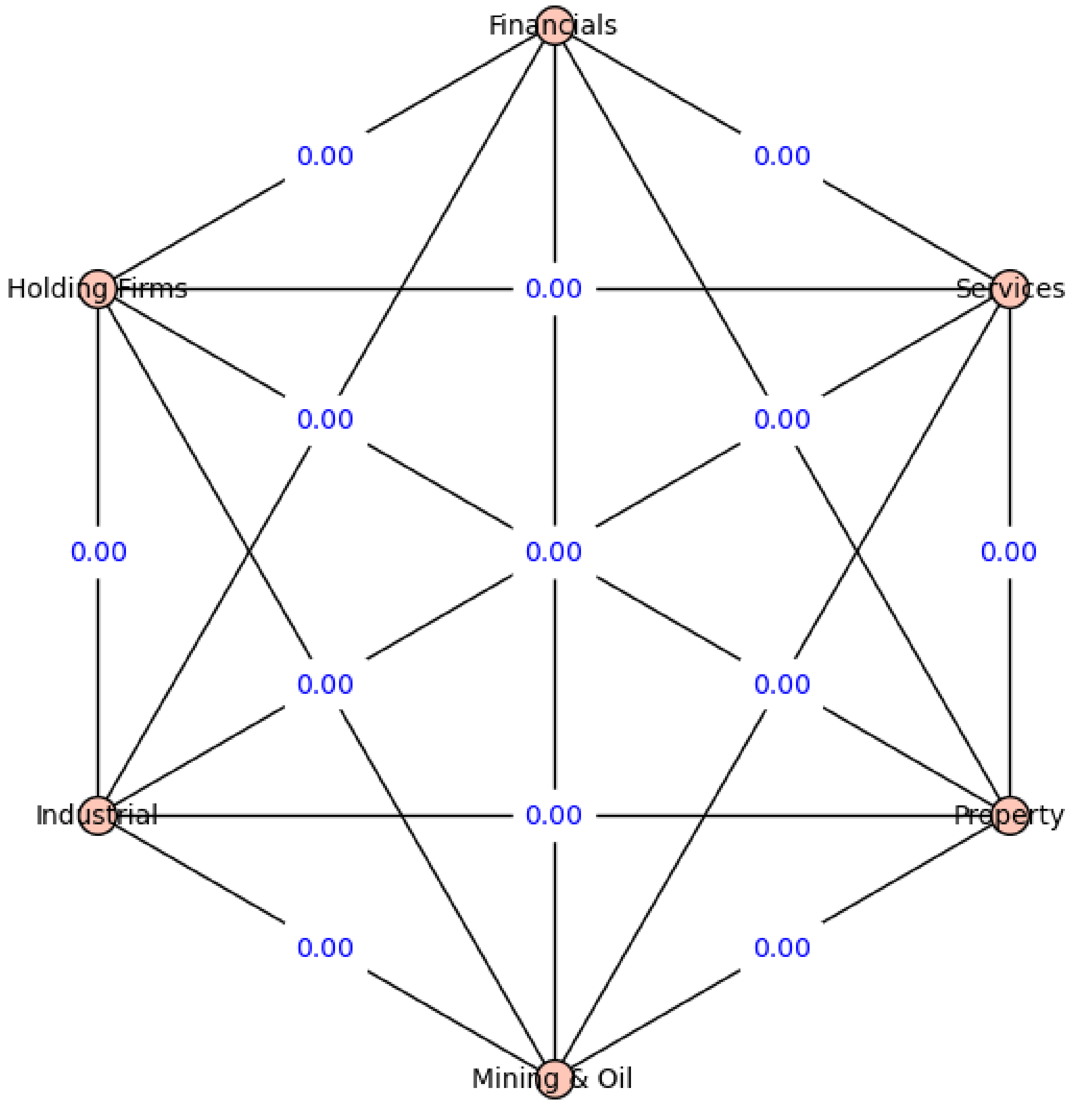
2. K6 complete graph where edge weights is the batch b correlation coefficient

3. K6 complete graph where edge weights is the distance between two vertices with batch a correlation coefficient as a predictor variable

4. K6 complete graph where edge weights is the distance between two vertices with

Figure 1

The baseline K6 graph with assumed initial weights of 0.00 for visualization purposes



batch b correlation coefficient as a predictor variable

5. Minimum Spanning Tree showing the Market Structures with the edge weights being the distance between two vertices with batch a correlation coefficient as a predictor variable

6. Minimum Spanning Tree showing the Market Structures with the edge weights being the distance between two vertices with batch b correlation coefficient as a predictor variable

MAIN DISCUSSION

Data Collection

Data collection will be conducted via manual encoding of tabled weekly closing prices data on Trading view. This will be the main data collection methodology of the exposition utilizing the stock tickers from the sector's compositions to make that aggregated node and its statistical makeup and traits.w is a web-based platform that provides advanced charting and analysis tools for traders and investors.It is of the most popular charting platforms, resources and , market data globally, serving millions of users in more than 100 countries (TradingView, 2026). the table bellow 2 specifies the all the PSE sectors.

Weighing the edges of the MST

The correlation coefficient is a statistical used to measure the strength and direction of relationship between two variables this degree of relationship can be expressed as

$$p_{ij} = \frac{< Y_i Y_j > - < Y_i >< Y_j >}{\sqrt{(< Y_i^2 > - < Y_i >^2)(< Y_j^2 > - < Y_j >^2)}}$$

Where i and j are numerical labels of stocks, $Y_i = lnP_i(t) - lnP_i(t-1)$ as the closure prices of stock i at time horizon t and a correlation matrix is to be made from these

Table 2

Specifications of PSE Sectoral Indices with Observations

Index	**Intervals**	**Batch**	**Start Date**	**End Date**	**n Observations**
Property	Weekly	First	07-01-2019	11-25-2019	21
		Second	12-02-2019	05-25-2020	26
Mining & Oil	Weekly	First	07-01-2019	11-25-2019	21
		Second	12-02-2019	05-25-2020	26
Industrial	Weekly	First	07-01-2019	11-25-2019	21
		Second	12-02-2019	05-25-2020	26
Financials	Weekly	First	07-01-2019	11-25-2019	21
		Second	12-02-2019	05-25-2020	26
Services	Weekly	First	07-01-2019	11-25-2019	21
		Second	12-02-2019	05-25-2020	26
Holding Firms	Weekly	First	07-01-2019	11-25-2019	21
		Second	12-02-2019	04-27-2020	26

numerical evaluation (Mantegna, 1999). Mantegna used stocks as the sets of data to be the evaluated but the exposition will use the six sectors of PSEI as the P_i and P_j on evaluating the correlation between them that will be the vital quantity in formulating the edges of the graphs between sector i and j and correlation matrix post evaluation.

A correlation coefficient is expected to return a value x such that: $-1 \leq x \leq 1$ where if x is positive it means two variables have a degree of positive linear correlation while if it is negative it means two variables have a degree of negative linear correlation. the nearer X is to -1 and 1 the stronger the linear relationship is (statstutor, 2013).

Categorization of Correlation Coefficient Scores

In order to determine the strength of the relationship of two variables from the computed correlation coefficient a interpretative table must be constructed to determine the threshold of each categories (statstutor, 2013)

Table 3

Interpretation of Correlation Coefficients

Range	Strength of Relationship
.00–.19	Very weak
.20–.39	Weak
.40–.59	Moderate
.60–.79	Strong
.80–1.0	Very strong

Euclidean Metric and the generalized distance function with the correlation value as its pre-image

The reference study used distance to assign weight to the edges and create a hierarchical organization of nodes the function the original study formulated was:

$$d(i, j) = \sqrt{2(1 - p_{ij})}$$

From that distance equation a distance matrix is to be created in order for the MST to be created connecting the nodes given. A MST was chosen to represent the structure due to allowing arrangements of nodes that provides the most relevant relationships of the nodes (Mantegna, 1999).

Kruskal's Algorithm

Kruskal's algorithm provides a way to create the MST from a distance matrix. It process the edges in the order of the weight of their values taking and coloring for the MST each edges that does not form a cycle with past edges added stopping at the last edge that could be possibly added the colored edges forms a forest of trees that will overtime form to a single tree the MST (Sedgewick & Wayne, 2011). This will be the algorithm that will be used to form the MST in the exposition.

Results

Batch A

As observed in 4 most of the sector pairs exhibit moderate to weak correlation implying with several outliers such as: Industrial and Holding Firms Pairing being at 0.700 within the range of Strong relationship as shown at 3 this shows that during a relatively stable market condition before the Covid-19 pandemic most of the financial markets have their closing prices move independently of one another excluding the outlier Holding Firms - Industrial pairing that has a structural relationship with one another.

The Minimum Spanning Tree for Batch A the relatively stable market condition before Covid-19 we can see that each vertices has this much degrees: The table 6 shows the Industrial Sector was a critical sector of the economy that has links to Holding Firms, Services and Property a negative event on the industrial sector will likely produce reactive negative events on other sectors linked to it to the tree. The total weight of the tree is 5.014 showing the market at this time frame is resilient a single vertex failing and influencing other vertices.

Batch B

As observed in 7 all of the sector pairings did show strong correlation with one another as shown at 3 this shows that during the Covid-19 pandemic the financial markets experienced a loss of independence in the closing price movements showing how the economic slowdown in one sector caused ripple effects with one another during that time frame.

The Minimum Spanning Tree for Batch B the market condition under Covid-19 has been observed to show the collapse of closing price movements independence of most sectors of the economy as shown in 9 with the convergence of vertices to Holding Firms and general shrinkage of th distances in all pairs. The structural risks posed by Covid-19

Table 4

Correlation Matrix First Batch

	Property	**Mining & Oil**	**Financials**	**Holding Firms**	**Industrial**	**Services**
Property	1.000	0.290	0.400	0.427	0.448	0.206
Mining & Oil	0.290	1.000	0.384	0.361	0.227	0.061
Financials	0.400	0.384	1.000	0.353	0.239	0.323
Holding Firms	0.427	0.361	0.353	1.000	0.700	0.493
Industrial	0.448	0.227	0.239	0.700	1.000	0.515
Services	0.206	0.061	0.323	0.493	0.515	1.000

Note. Values are Pearson correlation coefficients.

Figure 2

The K6 weighted graph with the weights being E_{ij} for visualization purposes of batch A

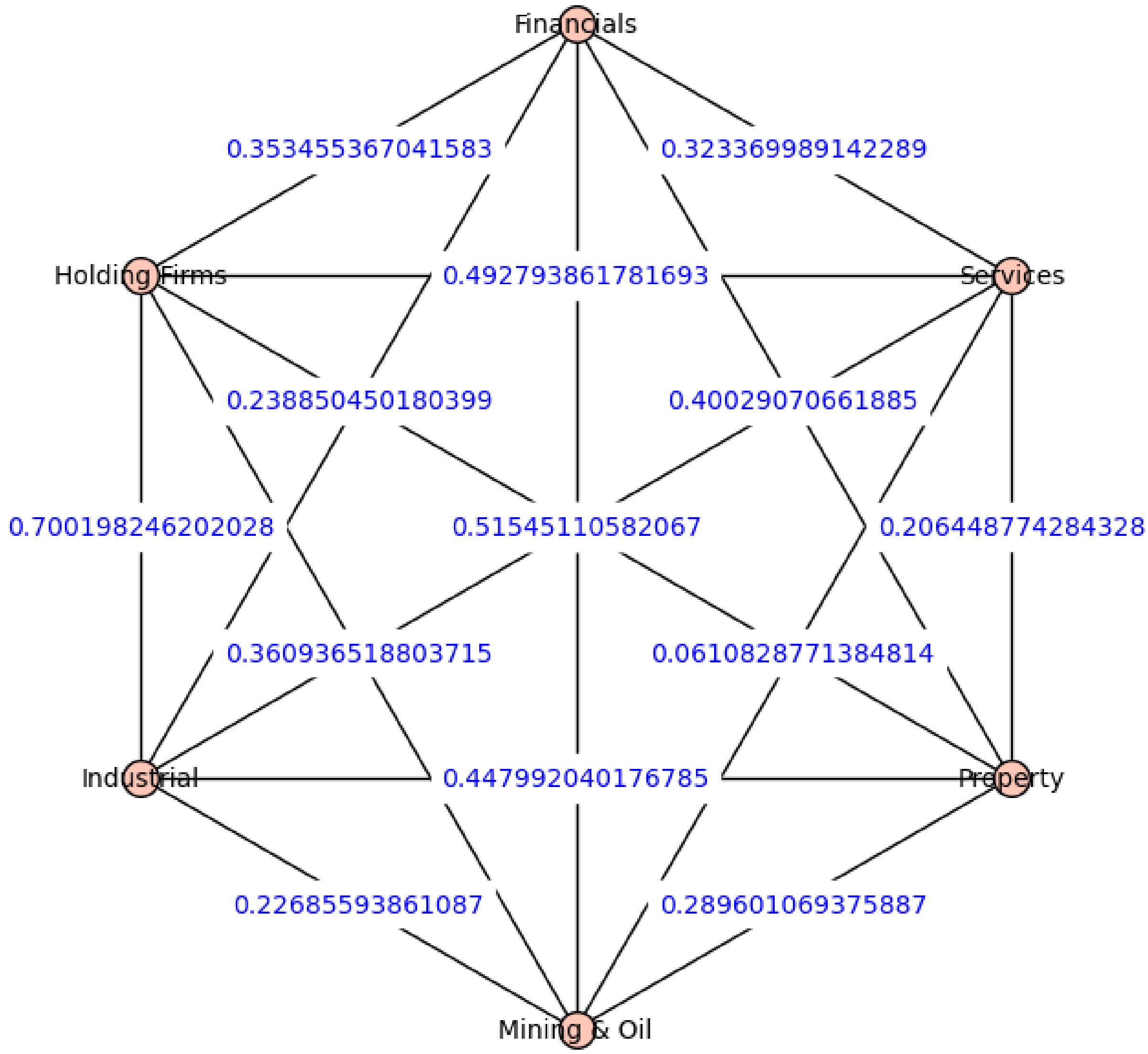


Table 5

Distance Matrix First Batch

	Property	**Mining & Oil**	**Financials**	**Holding Firms**	**Industrial**	**Services**
Property	0.000	1.192	1.095	1.070	1.051	1.260
Mining & Oil	1.192	0.000	1.110	1.131	1.243	1.370
Financials	1.095	1.110	0.000	1.137	1.234	1.163
Holding Firms	1.070	1.131	1.137	0.000	0.774	1.007
Industrial	1.051	1.243	1.234	0.774	0.000	0.984
Services	1.260	1.370	1.163	1.007	0.984	0.000

Note. Pairwise distances between sectors; diagonal entries are zero.

Figure 3

The K6 weighted graph with the weights being E_{ij} for visualization purposes

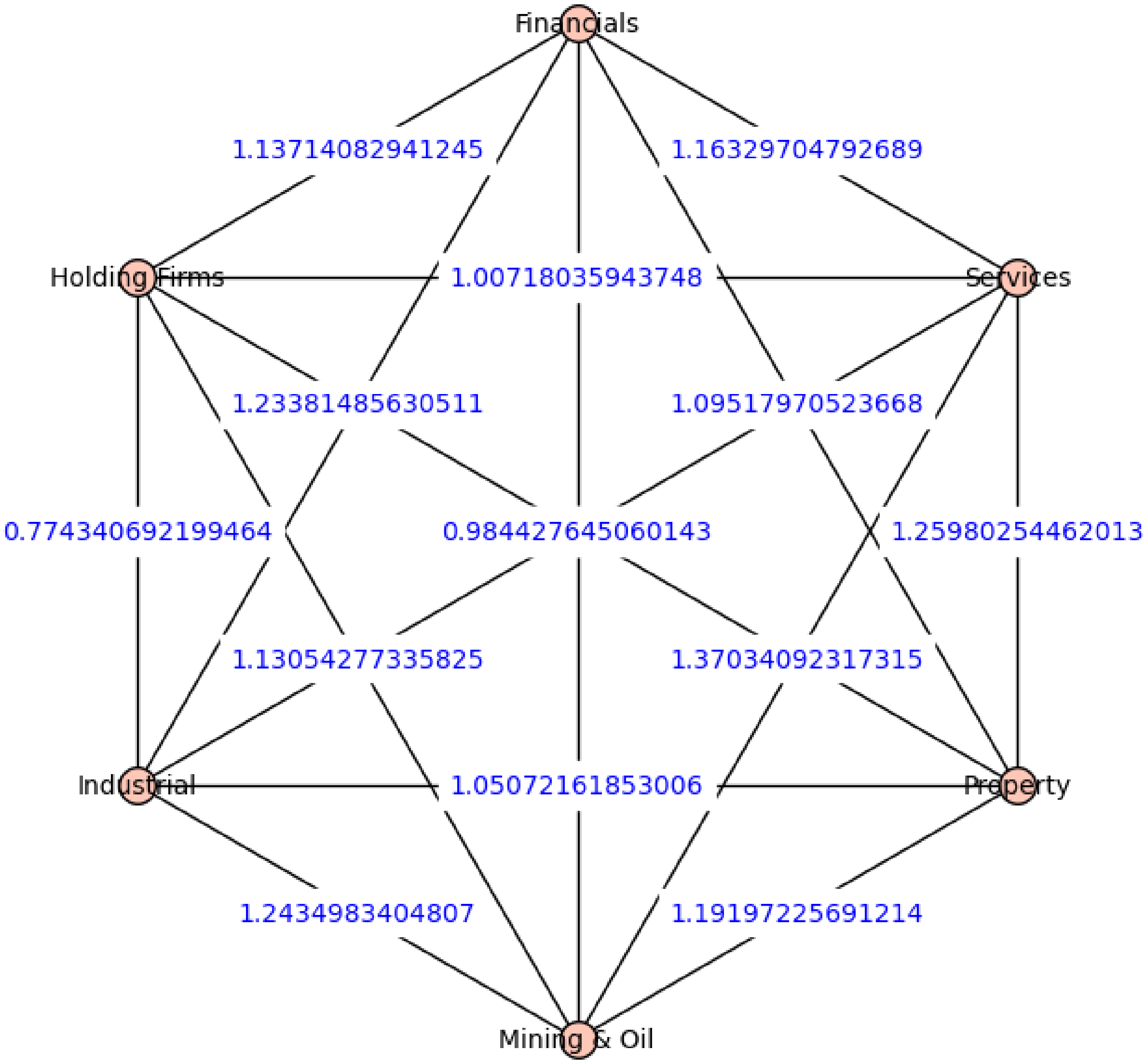


Table 6

Batch A Node Degrees and Connections

Node	**Degree**	**Connections**
Industrial	3	Holding; Services; Property
Property	2	Industrial; Financials
Financials	2	Property; Mining & Oil
Holding Firms	1	Industrial
Services	1	Industrial
Mining & Oil	1	Financials

Note. Degree equals the number of connections.

Table 7

Correlation Matrix Second Batch

	Property	**Mining & Oil**	**Financials**	**Holding Firms**	**Industrial**	**Services**
Property	1.000	0.868	0.844	0.920	0.807	0.873
Mining & Oil	0.868	1.000	0.793	0.834	0.813	0.877
Financials	0.844	0.793	1.000	0.898	0.792	0.838
Holding Firms	0.920	0.834	0.898	1.000	0.830	0.894
Industrial	0.807	0.813	0.792	0.830	1.000	0.811
Services	0.873	0.877	0.838	0.894	0.811	1.000

Note. Values are Pearson correlation coefficients.

Table 8

Distance Matrix Second Batch

	Property	**Mining & Oil**	**Financials**	**Holding Firms**	**Industrial**	**Services**
Property	0.000	0.513	0.559	0.400	0.621	0.504
Mining & Oil	0.513	0.000	0.644	0.576	0.612	0.496
Financials	0.559	0.644	0.000	0.451	0.645	0.570
Holding Firms	0.400	0.576	0.451	0.000	0.584	0.461
Industrial	0.621	0.612	0.645	0.584	0.000	0.615
Services	0.504	0.496	0.570	0.461	0.615	0.000

Note. Pairwise distances between sectors; diagonal entries are zero.

Figure 4

The Minimum Spanning Tree of Batch A

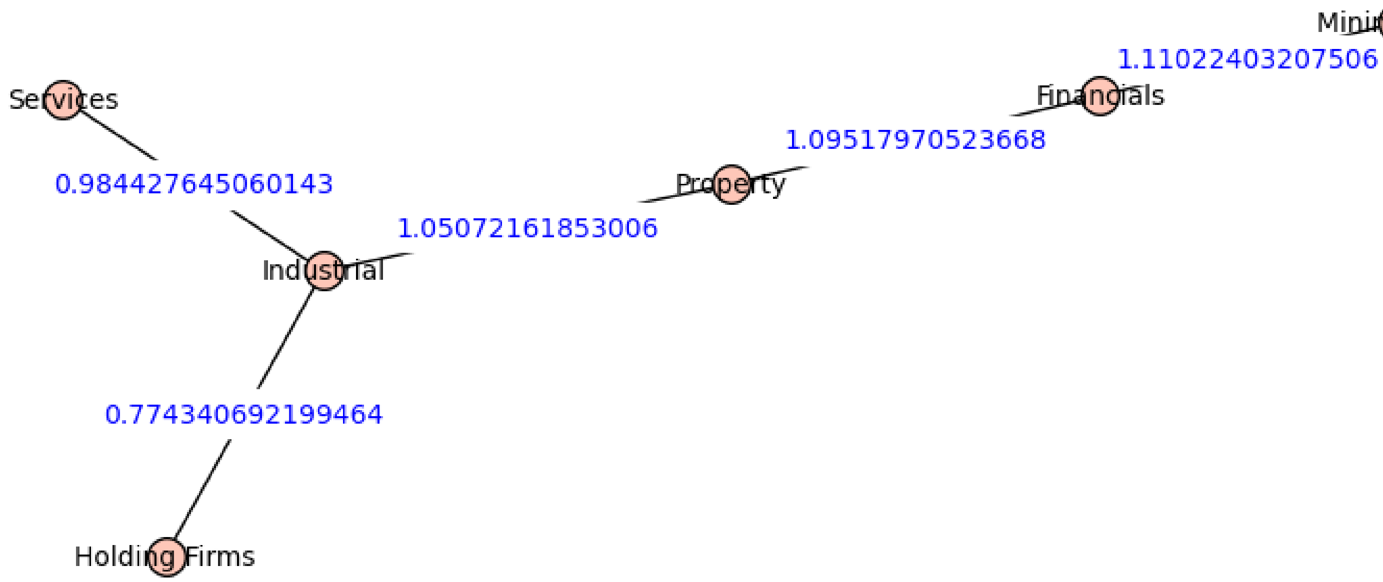


Table 9

Batch B Node Degrees and Connections

Node	Degree	Connections
Holding Firms	4	Property; Financials; Services; Industrial
Services	2	Holding; Mining & Oil
Property	1	Holding
Financials	1	Holding
Mining & Oil	1	Services
Industrial	1	Holding

Note. Degree equals the number of connections.

Figure 5

The K6 weighted graph with the weights being E_{ij} for visualization purposes of batch B

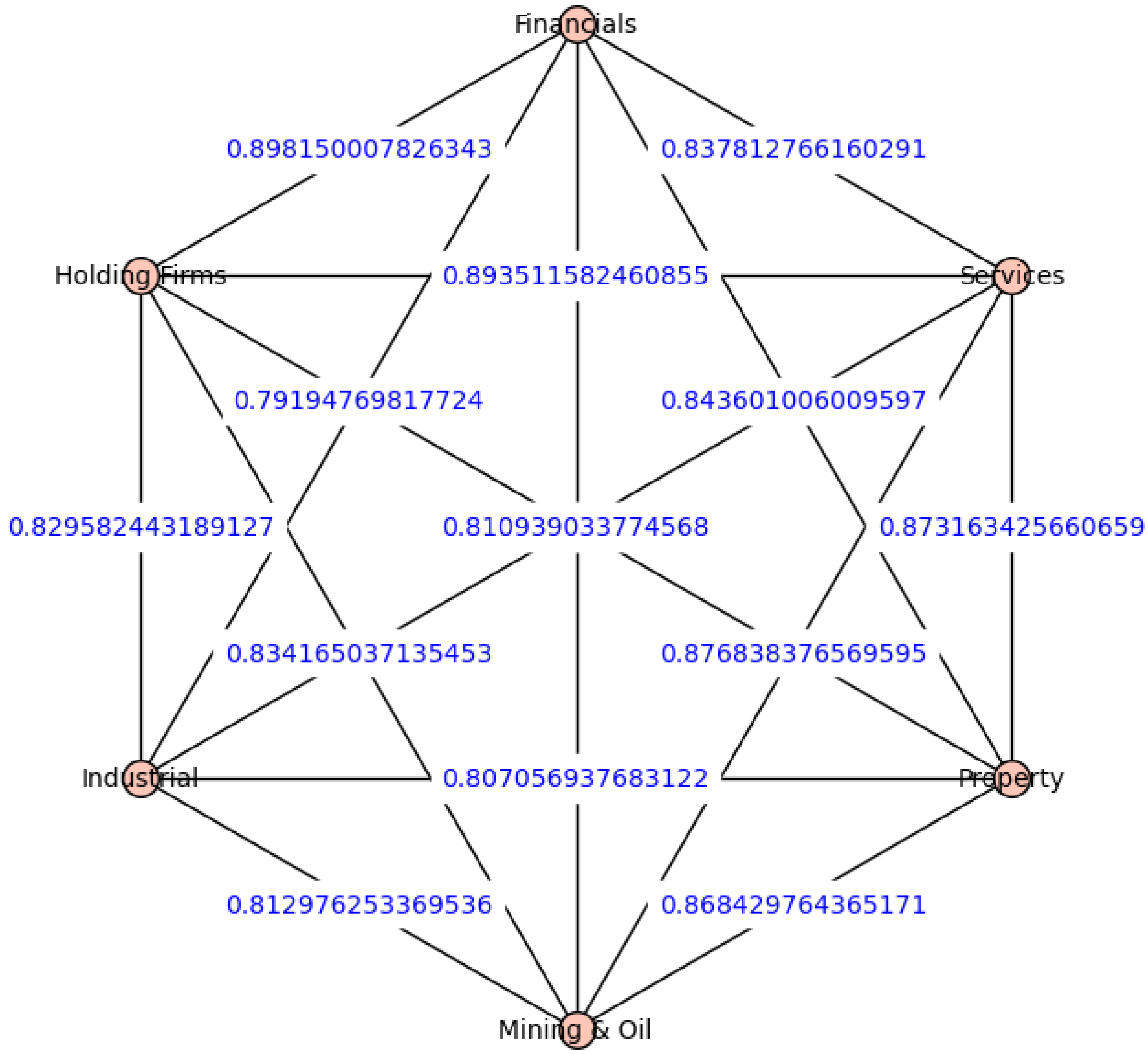
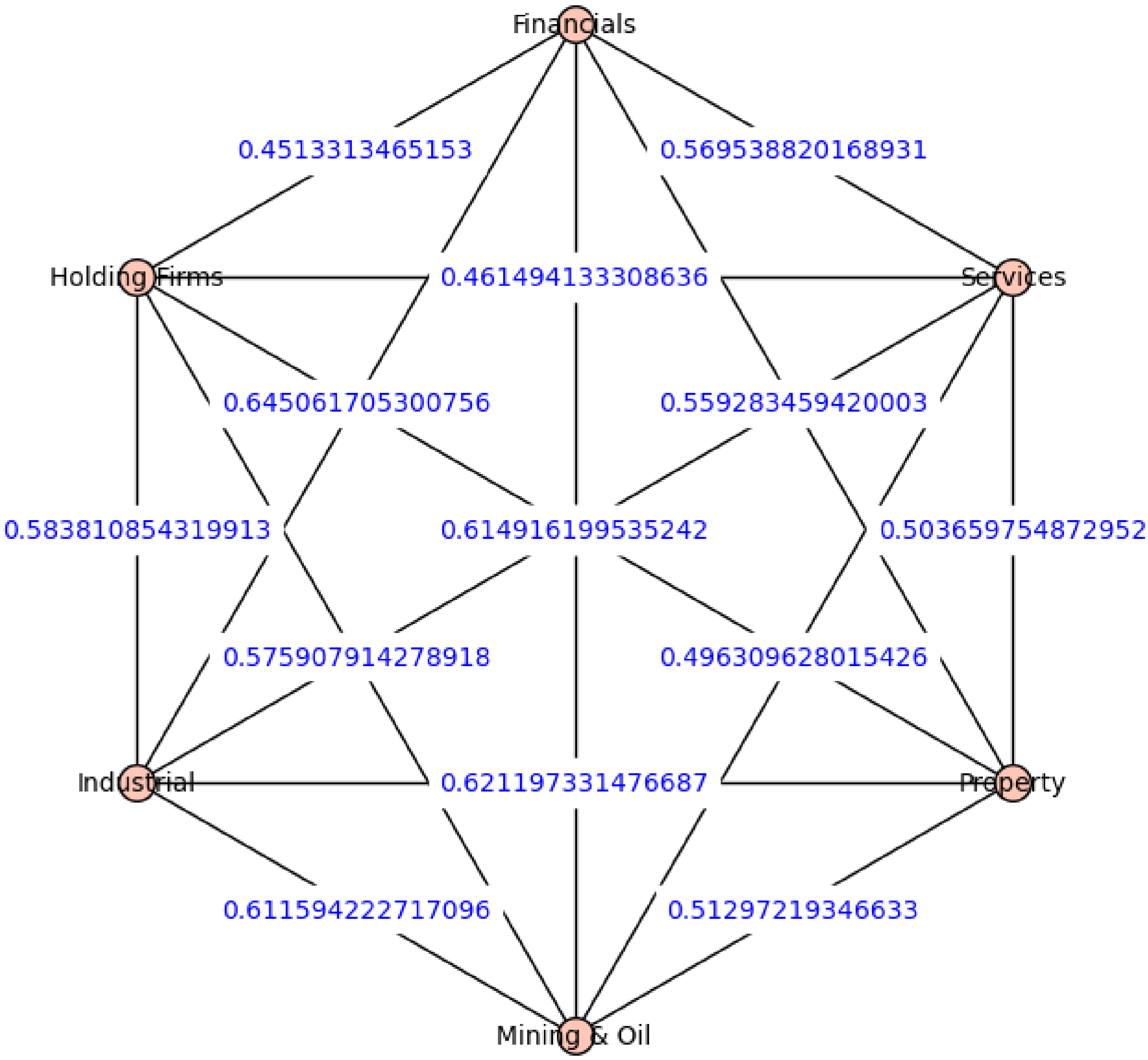


Figure 6

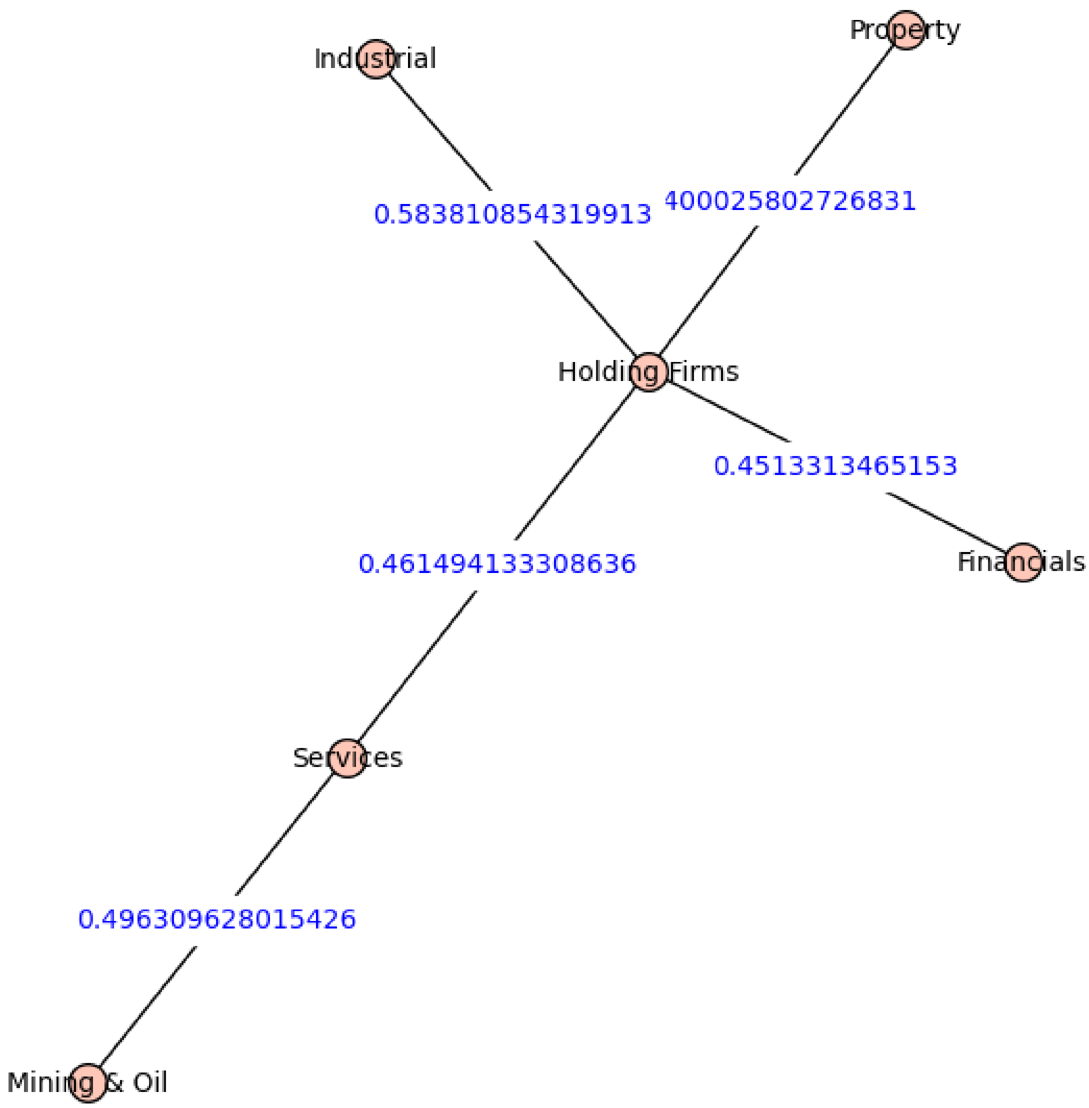
The K6 weighted graph with the weights being E_{ij} for visualization purposes



on the market exhibits the structural risks as the market conditions have changed from the economy before more loosely anchored to the Industrial Sector as shown in 4 as it is linked to Industrial - Holding Firms, Industrial - Services, and Holding Firms - Property. Then during the Covid-19 pandemic the hub of the market shifted to Holding Firms with 4 sectors highly connected to it as shown by a lower distance to the Holding Firms Vertex. The total weight of the Tree from 5.014 to 2.392 is an observation of how the pandemic coupled the sectors of the economy in the downturn.

Figure 7

The Minimum Spanning Tree of Batch B



Discussion

Difference in the Weight of the Pairs from Batch A to Batch B

As observed in 10 the sector pairings have generally observed a change in the statistical properties of the macro-aggregates towards a more correlated closing prices movements. To attain which sector parings had the most dramatic shift among the observations we will use the third quartile to give us the sector pairs that had the most dramatic changes.

The quartile formula is used to divide groups into quarters the quartile formula to attain the third quartile is:

$$\frac{3(n+1)}{4} = thresholdterm$$

and allow us to see the group of correlation deltas that has shifted the most dramatically within the whole group (SIES College of Commerce and Economics, 2024)

which give us the $12th$ term at 10 which is: $correlationdelta \geq 0.578828694989285$ and the sector pairs that met this threshold are: Property - Mining & Oil, Mining & Oil - Industrial, Property - Services, and Mining & Oil - Services. This observations might imply that Mining & oil is particularly vulnerable sector when market regimes shifts. And satisfies the objective 1's goal of finding the sector pairs that have the most different correlations from batch A to batch B.

Structural Backbone of the sectors

As observed in the 6 and 9 the Holding Firms - Industrial sector pairing is the Structural hub that many sectors are structurally linked. This highly resilient sector pairings might reflect a more structural couplings of the two as holding firms had many assets in the industrial sectors of the country.

Table 10

Correlation delta stability ranking

Ranking	$\mathbf{Y_i}$	$\mathbf{Y_j}$	**1st**	**2nd**	**Correlation delta (2nd - 1st)**
1	Holding Firms	Industrial	0.7001982460202	0.829582443189127	0.129384196987099
2	Industrial	Services	0.51545110582067	0.610939033774568	0.0954879279538896
3	Property	Industrial	0.447992040176785	0.807056937683122	0.3590648975060337
4	Holding Firms	Services	0.492793861781693	0.893511582460856	0.400717720679163
5	Mining & Oil	Financials	0.383701299301501	0.792351453124565	0.408650153823064
6	Property	Financials	0.400209706618653	0.8443600090597	0.444150302441047
7	Mining & Oil	Holding Firms	0.360936518803715	0.834165037135454	0.473228518331739
8	Property	Holding Firms	0.427268116077046	0.91999876756378	0.492721652499933
9	Financials	Services	0.323369899142289	0.837812766160292	0.514442866017993
10	Financials	Holding Firms	0.353455367041858	0.898150007826343	0.54469464078476
11	Financials	Industrial	0.2238385004573	0.779147698117221	0.555309197659921
12	Property	Mining & Oil	0.289601069375887	0.868429764365171	0.578828694989284
13	Mining & Oil	Industrial	0.226559386810869	0.812976253396536	0.586120134758668
14	Property	Services	0.206448774428638	0.873611342066599	0.667162567637961
15	Mining & Oil	Services	0.0610828771384807	0.876838376569595	0.815755499431115

Observation on the Distances between the Batch A and Batch B and how Covid-19 affected the financial market

We can see a dramatic contraction of distances between the the first batch batch A and second batch batch B as all distance delta is negative showing that the $d_{batchA}(i_{batchA}, j_{batch_a}) > d_{batch_b}(i_{batch_B}, j_{batch_B})$ indicating that the closing price movements of each sector lost a degree of statistical independence and became more correlated with one another as total distance of 4 is 5.014 and the distance of 7 is 2.392 showing that the Covid-19 pandemic and ECQ has disrupted the financial markets gravely.

Notable Sectors that has exhibited stability here are: Holding Firms - Industrial as with delta of -0.190 in distance from one another in MST of the same edge is remarked as the most resilient edge despite the market shock.With Industrial - Services being the next most stable edge with a distance delta of -0.369 followed by Property - Industrial and Mining & Oil - Financials bearing a delta that is $-0.399 \leq x \leq -0.500$ indicating that sector pairings involving the industrial sector maintained relatively stable structural distances across both market regimes implying a level of structural resilience that Covid 19 pandemic failed to disrupt significantly.

REFERENCES

Bangko Sentral ng Pilipinas. (2023). *Bsp unbound: Central banking and the post-pandemic economy*. https://www.bsp.gov.ph/Media_And_Research/Publications/BSP_Unbound.pdf

Cacnio, F. C. Q. (2020). *The impact of monetary policy on the economy* (Working Paper Series No. 2020-05). Bangko Sentral ng Pilipinas. https://www.bsp.gov.ph/Media_And_Research/WPS/WPS202005.pdf

Danko, J., & Šoltés, V. (2018). Portfolio creation using graph characteristics. *Investment Management and Financial Innovations*, *15*(1), 180–189. https://doi.org/10.21511/imfi.15(1).2018.16

Table 11

Distance delta stability ranking

Ranking	**Y_i**	**Y_j**	**1st**	**2nd**	**Distance delta (2nd - 1st)**
1	Holding Firms	Industrial	0.774340692199464	0.583810854319913	-0.190529837879551
2	Industrial	Services	0.984427645060143	0.614916199535242	-0.369511445524902
3	Property	Industrial	1.05072161853006	0.621197331476687	-0.429524287053374
4	Mining & Oil	Financials	1.11022403207506	0.64415601967755	-0.466067922307509
5	Property	Financials	1.09519770523668	0.559283459420003	-0.535896245816677
6	Holding Firms	Services	1.00718035934783	0.461494133308863	-0.545686226038967
7	Mining & Oil	Holding Firms	1.13054277335825	0.575907914278905	-0.554634859079345
8	Financials	Industrial	1.23381485630511	0.645061750300756	-0.588753151004355
9	Financials	Services	1.16329704891162	0.56953882016893	-0.593758227715689
10	Mining & Oil	Industrial	1.24349894408407	0.611594222717096	-0.631904117366976
11	Property	Holding Firms	1.07026341066184	0.400082200872802	-0.670181209789038
12	Property	Mining & Oil	1.19197225629161	0.51297219346633	-0.679000062845812
13	Financials	Holding Firms	1.13054277335825	0.451331465311635	-0.679211226846593
14	Property	Services	1.25980254446201	0.503659874275192	-0.756142670186818
15	Mining & Oil	Services	1.37034092317315	0.614916199535242	-0.755524723637907

Hodula, M., Pfeifer, L., & Janků, J. (2022, September). *The effect of structural risks on financial downturns* (Working Paper No. 138). European Systemic Risk Board (ESRB). Frankfurt am Main, Germany. https://www.esrb.europa.eu/pub/pdf/wp/esrb.wp138~482836b9ef.en.pdf

Mantegna, R. N. (1999). Hierarchical structure in financial markets. *The European Physical Journal B: Condensed Matter and Complex Systems*, *11*(1), 193–197. https://doi.org/10.1007/s100510050929

Sedgewick, R., & Wayne, K. (2011). Minimum spanning trees. https://algs4.cs.princeton.edu/43mst/

SIES College of Commerce and Economics. (2024). Quartiles [Educational resource on calculating quartiles].

statstutor. (2013). Pearson's correlation coefficient [Accessed: March 18, 2026]. https://www.statstutor.ac.uk/resources/uploaded/pearsons.pdf

The Philippine Stock Exchange, Inc. (2026). The philippine stock exchange. https://www.pse.com.ph/

TradingView. (2026). Tradingview — track all markets [Financial charting platform and social network for traders and investors].

APPENDIX

Table 12

Table 1
PSE Property Index — weekly closing prices and log values

Date	**Closing price**	$\ln p_{t-1}$	$\ln p_t$	$\ln p_t - \ln p_{t-1}$
7/1/2019	4,379.75	8.384746924		
7/8/2019	4,423.93	8.394787366	8.394787366	0.010038641
7/15/2019	4,425.37	8.394702296	8.394702296	-0.000085070
7/22/2019	4,347.09	8.377761898	8.377761898	-0.016940398

Continued on next page

Table 1 (continued)

Date	**Closing price**	$\ln p_{t-1}$	$\ln p_t$	$\ln p_t - \ln p_{t-1}$
7/29/2019	4,103.24	8.319187811	8.319187811	-0.058574087
8/5/2019	4,100.02	8.318747131	8.318747131	-0.053189048
8/12/2019	4,036.97	8.303249690	8.303249690	-0.015497441
8/19/2019	4,016.69	8.298213434	8.298213434	-0.005036255
8/26/2019	3,998.00	8.293549515	8.293549515	-0.004663919
9/2/2019	4,050.97	8.306711638	8.306711638	0.013162123
9/9/2019	4,064.08	8.309942699	8.309942699	0.003231061
9/16/2019	3,985.76	8.290483288	8.290483288	-0.019459410
9/23/2019	4,068.62	8.311059179	8.311059179	0.020575891
9/30/2019	4,010.90	8.296770909	8.296770909	-0.014288270
10/7/2019	4,129.68	8.325955250	8.325955250	0.029184340
10/14/2019	4,177.25	8.337408414	8.337408414	0.011453165
10/21/2019	4,166.29	8.334781231	8.334781231	-0.002627184
10/28/2019	4,172.29	8.336220325	8.336220325	0.001439094
11/4/2019	4,170.89	8.335884745	8.335884745	-0.000335579
11/11/2019	4,119.84	8.323569558	8.323569558	-0.012315187
11/18/2019	4,033.22	8.302320343	8.302320343	-0.021249215
11/25/2019	4,043.60	8.304890688	8.304890688	0.002570345
12/2/2019	4,099.88	8.318712960	8.318712960	0.013822272
12/9/2019	4,193.94	8.341395881	8.341395881	0.022682922
12/16/2019	4,139.54	8.328339950	8.328339950	-0.013055932
12/23/2019	4,154.52	8.331952177	8.331952177	0.003612228
12/30/2019	4,173.67	8.336551000	8.336551000	0.004598823
1/6/2020	4,112.34	8.321747440	8.321747440	-0.014803560
1/13/2020	4,035.78	8.302954871	8.302954871	-0.018792569

Continued on next page

Table 1 (continued)

Date	**Closing price**	$\ln p_{t-1}$	$\ln p_t$	$\ln p_t - \ln p_{t-1}$
1/20/2020	3,962.32	8.284585017	8.284585017	-0.018369854
1/27/2020	3,838.74	8.252899467	8.252899467	-0.031685550
2/3/2020	4,036.80	8.303207578	8.303207578	0.050308111
2/10/2020	3,928.65	8.276051109	8.276051109	-0.027156469
2/17/2020	3,944.75	8.280140860	8.280140860	0.004089751
2/24/2020	3,633.18	8.197863549	8.197863549	-0.082277311
3/2/2020	3,684.87	8.211990553	8.211990553	0.014127004
3/9/2020	2,985.40	8.001488987	8.001488987	-0.210501566
3/16/2020	2,468.33	7.811297128	7.811297128	-0.190191859
3/23/2020	2,781.15	7.930619754	7.930619754	0.119322626
3/30/2020	2,824.16	7.945966219	7.945966219	0.015346465
4/6/2020	2,838.40	7.950995757	7.950995757	0.005029538
4/13/2020	2,963.69	7.994190359	7.994190359	0.043194602
4/20/2020	2,715.46	7.906716647	7.906716647	-0.087473712
4/27/2020	2,921.92	7.979996179	7.979996179	0.073279532
5/4/2020	2,934.97	7.984452511	7.984452511	0.004456332
5/11/2020	2,838.40	7.950995757	7.950995757	-0.033456754
5/18/2020	2,815.65	7.942948384	7.942948384	-0.008047373
5/25/2020	2,785.90		7.932326224	-0.010622216

Note. Specifications: name = PSE Property Index; intervals = weekly; batch = first (7/1/2019–11/25/2019) and second (12/2/2019–5/25/2020). Hyperlink: https://www.tradingview.com/chart/?symbol=PSE%3APRO.

Table 13

Table 1
PSE Mining & Oil Index — weekly closing prices and log values

Date	**Closing price**	$\ln p_{t-1}$	$\ln p_t$	$\ln p_t - \ln p_{t-1}$
2019-07-01	7,439.29	8.914530693		
2019-07-08	7,437.16	8.914244335	8.914244335	-0.000286359
2019-07-15	8,126.27	9.002857303	9.002857303	0.088612968
2019-07-22	7,870.97	8.970936587	8.970936587	-0.031920716
2019-07-29	8,101.47	8.999800806	8.999800806	0.028864219
2019-08-05	8,152.22	9.006045562	9.006045562	0.006244756
2019-08-12	8,051.19	8.993575186	8.993575186	-0.012470376
2019-08-19	8,074.00	8.996404301	8.996404301	0.002829116
2019-08-26	8,250.52	9.018031508	9.018031508	0.021627206
2019-09-02	9,269.45	9.134479326	9.134479326	0.116447818
2019-09-09	9,591.46	9.168628398	9.168628398	0.034149073
2019-09-16	9,375.03	9.145805051	9.145805051	-0.022823347
2019-09-23	8,968.40	9.101462567	9.101462567	-0.044342484
2019-09-30	8,843.07	9.087389380	9.087389380	-0.014073186
2019-10-07	9,178.36	9.124603818	9.124603818	0.037214438
2019-10-14	8,907.35	9.094632058	9.094632058	-0.029971761
2019-10-21	9,209.23	9.127961521	9.127961521	0.033329463
2019-10-28	9,216.52	9.128752805	9.128752805	0.000791284
2019-11-04	9,092.61	9.115217275	9.115217275	-0.013535530
2019-11-11	8,742.14	9.075910290	9.075910290	-0.039306985
2019-11-18	8,175.43	9.008888594	9.008888594	-0.067021696
2019-11-25	8,054.07	8.993932833	8.993932833	-0.014955761
2019-12-02	7,517.51	8.924990245	8.924990245	-0.068942588

Continued on next page

Table 1 (continued)

Date	**Closing price**	$\ln p_{t-1}$	$\ln p_t$	$\ln p_t - \ln p_{t-1}$
2019-12-09	7,521.46	8.925515547	8.925515547	0.000525302
2019-12-16	7,586.74	8.934157266	8.934157266	0.008641719
2019-12-23	8,091.98	8.998628727	8.998628727	0.064471461
2019-12-30	8,020.76	8.989788459	8.989788459	-0.008840267
2020-01-06	8,107.64	9.000562106	9.000562106	0.010773647
2020-01-13	8,143.53	9.004979026	9.004979026	0.004416920
2020-01-20	7,968.19	8.983212644	8.983212644	-0.021766382
2020-01-27	7,663.26	8.944192760	8.944192760	-0.039019885
2020-02-03	7,383.47	8.906998997	8.906998997	-0.037193763
2020-02-10	7,058.64	8.862007677	8.862007677	-0.044991319
2020-02-17	7,086.90	8.866003288	8.866003288	0.003995611
2020-02-24	6,451.44	8.772058641	8.772058641	-0.093944648
2020-03-02	6,281.71	8.745397515	8.745397515	-0.026661125
2020-03-09	4,727.81	8.461217372	8.461217372	-0.284180143
2020-03-16	3,745.47	8.228302389	8.228302389	-0.232914983
2020-03-23	4,102.47	8.319344510	8.319344510	0.091042122
2020-03-30	4,261.87	8.357463310	8.357463310	0.038118800
2020-04-06	4,320.96	8.371232879	8.371232879	0.013769569
2020-04-13	4,688.22	8.452808258	8.452808258	0.081575380
2020-04-20	4,592.45	8.432168930	8.432168930	-0.020639329
2020-04-27	4,708.57	8.457139532	8.457139532	0.024970602
2020-05-04	4,520.85	8.416455308	8.416455308	-0.040684223
2020-05-11	4,505.48	8.413049713	8.413049713	-0.003405596
2020-05-18	4,465.27	8.404084962	8.404084962	-0.008964751
2020-05-25	4,362.72		8.380850995	-0.023233967

Note. Specifications: name = PSE Mining & Oil Index; intervals = weekly; batch = first (7/1/2019–11/25/2019) and second (12/2/2019–5/25/2020). Hyperlink: https://www.tradingview.com/chart/?symbol=PSE%3AM_O.

Table 14

Table 1
PSE Industrial Index — weekly closing prices and log values

Date	**Closing price**	$\ln p_{t-1}$	$\ln p_t$	$\ln p_t - \ln p_{t-1}$
2019-07-01	11,988.32	9.391688121	9.391688180	0.000000059
2019-07-08	11,791.25	9.375113010	9.375113010	-0.016575111
2019-07-15	11,467.74	9.367745750	9.367745750	-0.007367260
2019-07-22	11,704.70	9.347293155	9.347293155	-0.020452959
2019-07-29	11,412.13	9.342432104	9.342432104	-0.004861051
2019-08-05	10,804.27	9.287696705	9.287696705	-0.054735988
2019-08-12	10,844.32	9.291396772	9.291396772	0.003700014
2019-08-19	10,997.00	9.303777870	9.303777870	0.013981068
2019-08-26	11,126.30	9.317066954	9.317066954	0.011689167
2019-09-02	10,915.02	9.297895101	9.297895101	-0.019171853
2019-09-09	11,011.94	9.306738419	9.306738419	0.008840316
2019-09-16	10,938.94	9.300084178	9.300084178	-0.006651239
2019-09-23	10,556.72	9.264517903	9.264517903	-0.035566276
2019-09-30	10,477.26	9.256962473	9.256962473	-0.007555430
2019-10-07	10,677.38	9.275882764	9.275882764	0.018920291
2019-10-14	10,643.01	9.272658618	9.272658618	-0.003224146
2019-10-21	10,470.07	9.256275990	9.256275990	-0.016382628

Continued on next page

Table 1 (continued)

Date	Closing price	$\ln p_{t-1}$	$\ln p_t$	$\ln p_t - \ln p_{t-1}$
2019-10-28	10,470.94	9.256559008	9.256559008	0.000083095
2019-11-04	10,478.33	9.257064594	9.257064594	0.000705514
2019-11-11	10,082.34	9.218540658	9.218540658	-0.038523936
2019-11-18	9,911.62	9.201643095	9.201643095	-0.017075620
2019-11-25	9,781.29	9.188226656	9.188226656	-0.013263439
2019-12-02	9,555.43	9.164864827	9.164864827	-0.023361829
2019-12-09	9,722.08	9.182154877	9.182154877	0.017290050
2019-12-16	9,367.59	9.145011117	9.145011117	-0.037143760
2019-12-23	9,635.07	9.173164877	9.173164877	0.028153760
2019-12-30	9,684.14	9.178244733	9.178244733	0.005079856
2020-01-06	9,364.48	9.144679141	9.144679141	-0.033565593
2020-01-13	9,477.42	9.156667396	9.156667396	0.011988225
2020-01-20	9,728.72	9.182837584	9.182837584	0.026170188
2020-01-27	8,921.44	9.096212692	9.096212692	-0.086624891
2020-02-03	9,391.68	9.147579438	9.147579438	0.051366746
2020-02-10	8,936.64	9.097914914	9.097914914	-0.049664524
2020-02-17	8,987.08	9.103453280	9.103453280	0.005628367
2020-02-24	8,351.48	9.030194108	9.030194108	-0.073349173
2020-03-02	8,114.07	9.001354846	9.001354846	-0.028839261
2020-03-09	7,071.11	8.863772734	8.863772734	-0.137582112
2020-03-16	6,022.65	8.703282624	8.703282624	-0.160490110
2020-03-23	6,204.54	8.733065610	8.733065610	0.029753937
2020-03-30	6,433.97	8.769347076	8.769347076	0.036310515
2020-04-06	7,073.16	8.864062646	8.864062646	0.094715570
2020-04-13	7,566.82	8.931528512	8.931528512	0.067465506

Continued on next page

Table 1 (continued)

Date	**Closing price**	$\ln p_{t-1}$	$\ln p_t$	$\ln p_t - \ln p_{t-1}$
2020-04-20	7,209.48	8.883152106	8.883152106	-0.048376047
2020-04-27	7,474.93	8.919310060	8.919310060	0.036157955
2020-05-04	7,326.79	8.899292773	8.899292773	-0.020017288
2020-05-11	7,506.67	8.935472724	8.935472724	0.024552452
2020-05-18	7,304.81	8.896288328	8.896288328	-0.027258896
2020-05-25	7,311.09	8.897147625	8.897147625	0.000859297

Note. Specifications: name = PSE Industrial Index; intervals = weekly; batch = first (7/1/2019–11/25/2019) and second (12/2/2019–5/25/2020). Hyperlink: https://www.tradingview.com/chart/?symbol=PSE%3AIND.

Table 15

Table 1
PSE Financials Index — weekly closing prices and log values

Date	**Closing price**	$\ln p_{t-1}$	$\ln p_t$	$\ln p_t - \ln p_{t-1}$
2019-07-01	1,742.01	7.462794898	7.462794898	-
2019-07-08	1,788.79	7.489294693	7.489294693	0.026499795
2019-07-15	1,877.28	7.537579200	7.537579200	0.048284507
2019-07-22	1,848.49	7.522124369	7.522124369	-0.015454831
2019-07-29	1,861.40	7.529084172	7.529084172	0.006959803
2019-08-05	1,851.60	7.523805409	7.523805409	-0.005278763
2019-08-12	1,779.29	7.483969887	7.483969887	-0.039835722

Continued on next page

Table 1 (continued)

Date	**Closing price**	$\ln p_{t-1}$	$\ln p_t$	$\ln p_t - \ln p_{t-1}$
2019-08-19	1,827.12	7.510496236	7.510496236	0.026526348
2019-08-26	1,833.28	7.513861991	7.513861991	0.003365756
2019-09-02	1,804.17	7.497855931	7.497855931	-0.016006006
2019-09-09	1,832.62	7.513501916	7.513501916	0.015645885
2019-09-16	1,786.58	7.488058457	7.488058457	-0.025445349
2019-09-23	1,803.55	7.497512224	7.497512224	0.009453767
2019-09-30	1,805.59	7.498642868	7.498642868	0.001130644
2019-10-07	1,821.17	7.507234431	7.507234431	0.008591744
2019-10-14	1,841.26	7.518205999	7.518205999	0.010970698
2019-10-21	1,876.79	7.537318154	7.537318154	0.019112751
2019-10-28	1,918.40	7.559246784	7.559246784	0.021928630
2019-11-04	1,946.75	7.573916594	7.573916594	0.014669810
2019-11-11	1,920.85	7.560523075	7.560523075	-0.013393519
2019-11-18	1,874.13	7.535899831	7.535899831	-0.024623245
2019-11-25	1,841.80	7.518498633	7.518498633	-0.017401197
2019-12-02	1,874.61	7.536155917	7.536155917	0.017657283
2019-12-09	1,903.17	7.551276196	7.551276196	0.015120279
2019-12-16	1,843.16	7.519236769	7.519236769	-0.032094270
2019-12-23	1,863.65	7.530292209	7.530292209	0.011055440
2019-12-30	1,844.48	7.519956274	7.519956274	-0.010339535
2020-01-06	1,827.52	7.510715135	7.510715135	-0.009237538
2020-01-13	1,846.91	7.521269251	7.521269251	0.010554116
2020-01-20	1,810.37	7.501865623	7.501865623	-0.019982728
2020-01-27	1,724.93	7.452941749	7.452941749	-0.048344774
2020-02-03	1,791.09	7.490579652	7.490579652	0.037637903

Continued on next page

Table 1 (continued)

Date	**Closing price**	$\ln p_{t-1}$	$\ln p_t$	$\ln p_t - \ln p_{t-1}$
2020-02-10	1,738.95	7.461036762	7.461036762	-0.029542890
2020-02-17	1,738.92	7.461019510	7.461019510	-0.0000172519
2020-02-24	1,605.99	7.381495668	7.381495668	-0.079523842
2020-03-02	1,596.23	7.375399878	7.375399878	-0.006095790
2020-03-09	1,415.66	7.255351132	7.255351132	-0.120042845
2020-03-16	1,163.69	7.059351270	7.059351270	-0.195999863
2020-03-23	1,212.28	7.104374140	7.104374140	0.045022871
2020-03-30	1,183.78	7.076467987	7.076467987	-0.027906153
2020-04-06	1,157.37	7.059054669	7.059054669	-0.022652519
2020-04-13	1,235.72	7.119409075	7.119409075	0.065503606
2020-04-20	1,188.69	7.080607139	7.080607139	-0.038801936
2020-04-27	1,188.67	7.080590314	7.080590314	-0.0000168254
2020-05-04	1,148.08	7.045846261	7.045846261	-0.034744053
2020-05-11	1,129.40	7.029441797	7.029441797	-0.016404464
2020-05-18	1,102.50	7.005335607	7.005335607	-0.024106190
2020-05-25	1,176.42	7.070231208	7.070231208	0.064895600

Note. Specifications: name = PSE Financials Index; intervals = weekly; batch = first (7/1/2019–11/25/2019) and second (12/2/2019–5/25/2020). Hyperlink: https://www.tradingview.com/chart/?symbol=PSE%3AFIN.

Table 16

Table 1
PSE Services Index — weekly closing prices and log values

Date	**Closing price**	$\ln p_{t-1}$	$\ln p_t$	$\ln p_t - \ln p_{t-1}$
2019-07-01	1,695.66	7.435827325	7.435827325	-
2019-07-08	1,693.65	7.434641242	7.434641242	-0.001186082
2019-07-15	1,681.71	7.427566412	7.427566412	-0.007074830
2019-07-22	1,651.90	7.409681420	7.409681420	-0.017884992
2019-07-29	1,613.12	7.385925471	7.385925471	-0.023755949
2019-08-05	1,591.59	7.372488796	7.372488796	-0.013436675
2019-08-12	1,568.85	7.358098164	7.358098164	-0.014390632
2019-08-19	1,589.89	7.371420110	7.371420110	0.013321946
2019-08-26	1,623.75	7.392493568	7.392493568	0.021073458
2019-09-02	1,594.99	7.374622746	7.374622746	-0.017870822
2019-09-09	1,623.61	7.392407344	7.392407344	0.017784598
2019-09-16	1,588.84	7.370759469	7.370759469	-0.021647875
2019-09-23	1,544.02	7.342144684	7.342144684	-0.028614785
2019-09-30	1,510.64	7.320288681	7.320288681	-0.021856003
2019-10-07	1,526.32	7.330614888	7.330614888	0.010326207
2019-10-14	1,530.25	7.333186400	7.333186400	0.002571512
2019-10-21	1,531.53	7.334022515	7.334022515	0.000836115
2019-10-28	1,513.05	7.321882760	7.321882760	-0.012139755
2019-11-04	1,551.96	7.347273927	7.347273927	0.025391167
2019-11-11	1,547.47	7.344376618	7.344376618	-0.002897309
2019-11-18	1,549.70	7.345816643	7.345816643	0.001440025
2019-11-25	1,546.48	7.343736660	7.343736660	-0.002079983
2019-12-02	1,499.35	7.312786960	7.312786960	-0.030949700

Continued on next page

Table 1 (continued)

Date	**Closing price**	$\ln p_{t-1}$	$\ln p_t$	$\ln p_t - \ln p_{t-1}$
2019-12-09	1,525.19	7.329874271	7.329874271	0.017087312
2019-12-16	1,533.34	7.335203642	7.335203642	0.005329371
2019-12-23	1,531.10	7.333747171	7.333747171	-0.001456471
2019-12-30	1,547.27	7.344274737	7.344274737	0.010527565
2020-01-06	1,555.66	7.349655172	7.349655172	0.005380435
2020-01-13	1,564.02	7.355014709	7.355014709	0.005359537
2020-01-20	1,544.66	7.342559100	7.342559100	-0.012455608
2020-01-27	1,475.16	7.296521737	7.296521737	-0.046037363
2020-02-03	1,502.87	7.315318920	7.315318920	0.018797183
2020-02-10	1,440.30	7.272606704	7.272606704	-0.042712216
2020-02-17	1,444.81	7.275733104	7.275733104	0.003126399
2020-02-24	1,351.38	7.208885172	7.208885172	-0.066847932
2020-03-02	1,330.54	7.193340154	7.193340154	-0.015545018
2020-03-09	1,187.61	7.079698163	7.079698163	-0.113641991
2020-03-16	1,037.71	6.944771641	6.944771641	-0.134926522
2020-03-23	1,177.27	7.070953478	7.070953478	0.126181837
2020-03-30	1,188.90	7.080783789	7.080783789	0.009830311
2020-04-06	1,222.24	7.108440520	7.108440520	0.027656731
2020-04-13	1,294.84	7.166424140	7.166424140	0.057983620
2020-04-20	1,300.57	7.170557909	7.170557909	0.004133769
2020-04-27	1,373.82	7.225350460	7.225350460	0.054792551
2020-05-04	1,322.68	7.187415260	7.187415260	-0.037935200
2020-05-11	1,314.04	7.180861640	7.180861640	-0.006553620
2020-05-18	1,322.13	7.186999351	7.186999351	0.006137711
2020-05-25	1,362.44	7.217032489	7.217032489	0.030033138

Note. Specifications: name = PSE Services Index; intervals = weekly; batch = first (7/1/2019–11/25/2019) and second (12/2/2019–5/25/2020). Hyperlink: https://www.tradingview.com/chart/?symbol=PSE%3ASVC.

Table 17

Table 1
PSE Holding Firms Index — weekly closing prices and log values

Date	**Closing price**	$\ln p_{t-1}$	$\ln p_t$	$\ln p_t - \ln p_{t-1}$
2019-07-01	7,797.21	8.961521256		-
2019-07-08	7,758.23	8.956509949	8.956509949	-0.005011762
2019-07-15	7,942.92	8.980036232	8.980036232	0.023526278
2019-07-22	7,999.12	8.987086827	8.987086827	0.007050595
2019-07-29	7,901.63	8.974824334	8.974824334	-0.012262493
2019-08-05	7,630.42	8.939898156	8.939898156	-0.034926178
2019-08-12	7,727.00	8.952475968	8.952475968	0.012577812
2019-08-19	7,772.54	8.958352288	8.958352288	0.005876320
2019-08-26	7,922.86	8.977507518	8.977507518	0.019155230
2019-09-02	7,908.14	8.975647490	8.975647490	-0.001859618
2019-09-09	7,918.59	8.976968413	8.976968413	0.001320913
2019-09-16	7,849.10	8.968154167	8.968154167	-0.008814246
2019-09-23	7,723.44	8.952015127	8.952015127	-0.016139040
2019-09-30	7,551.58	8.929512105	8.929512105	-0.022503022
2019-10-07	7,670.38	8.945121424	8.945121424	0.015609319
2019-10-14	7,676.89	8.945969809	8.945969809	0.000848385
2019-10-21	7,745.04	8.954807917	8.954807917	0.008838108

Continued on next page

Table 1 (continued)

Date	**Closing price**	$\ln p_{t-1}$	$\ln p_t$	$\ln p_t - \ln p_{t-1}$
2019-10-28	7,830.30	8.965756077	8.965756077	0.010948159
2019-11-04	7,933.32	8.978826865	8.978826865	0.013070788
2019-11-11	7,810.42	8.963214006	8.963214006	-0.015612859
2019-11-18	7,756.93	8.956341942	8.956341942	-0.006872064
2019-11-25	7,630.43	8.939899505	8.939899505	-0.016442437
2019-12-02	7,750.82	8.955535897	8.955535897	0.015636392
2019-12-09	7,633.31	8.940276857	8.940276857	-0.015257040
2019-12-16	7,599.65	8.935857459	8.935857459	-0.004419398
2019-12-23	7,592.07	8.934859534	8.934859534	-0.000997925
2019-12-30	7,631.72	8.940068551	8.940068551	0.005209017
2020-01-06	7,635.45	8.940557181	8.940557181	0.000488630
2020-01-13	7,475.59	8.919389298	8.919389298	-0.021167883
2020-01-20	7,348.36	8.902232424	8.902232424	-0.017156874
2020-01-27	6,900.31	8.839321632	8.839321632	-0.062910792
2020-02-03	7,146.01	8.874309410	8.874309410	0.034987778
2020-02-10	7,051.97	8.861062318	8.861062318	-0.013247092
2020-02-17	7,218.98	8.884468948	8.884468948	0.023406630
2020-02-24	6,627.08	8.798919579	8.798919579	-0.085549369
2020-03-02	6,612.26	8.796680750	8.796680750	-0.002238829
2020-03-09	5,673.65	8.643587911	8.643587911	-0.153092839
2020-03-16	4,602.06	8.434259330	8.434259330	-0.209328581
2020-03-23	5,142.11	8.545218761	8.545218761	0.110959431
2020-03-30	5,265.52	8.568935185	8.568935185	0.023716425
2020-04-06	5,510.79	8.614463268	8.614463268	0.045528083
2020-04-13	5,731.72	8.653770974	8.653770974	0.039307706

Continued on next page

Table 1 (continued)

Date	**Closing price**	$\ln p_{t-1}$	$\ln p_t$	$\ln p_t - \ln p_{t-1}$
2020-04-20	5,361.78	8.587051251	8.587051251	-0.066719723
2020-04-27	5,544.29	8.620523848	8.620523848	0.033472597
2020-05-04	5,532.27	8.618353499	8.618353499	-0.002170350
2020-05-11	5,437.09	8.600999233	8.600999233	-0.017354266
2020-05-18	5,540.44	8.619829181	8.619829181	0.018829948
2020-05-25	5,946.52	8.690561453	8.690561453	0.070732272

Note. Specifications: name = PSE Holding Firms Index; intervals = weekly; batch = first (7/1/2019–11/25/2019) and second (12/2/2019–5/25/2020). Hyperlink: https://www.tradingview.com/chart/?symbol=PSE%3AHDG.